

Standard form



1 Express the following equations in standard form.

a $y = 2x - 5$

b $y = \frac{2x}{5} + 4$

c $y = 6x - 5$

d $y = \frac{6x}{5} - 6$

e $y = \frac{-x}{-6}$

f $y - 5x = 5$

2 Luke wanted to rewrite $y = \frac{9x}{10} + 4$ in standard form. His steps are as follows:

1 $y = \frac{9x + 40}{10}$

2 $10y = 9x + 40$

3 $9x + 10y = 40$

State which step Luke made a mistake.

3 For each of the following equations of lines:

a Solve for x -value of the x -intercept of the line.

b Solve for y -value of the y -intercept of the line.

c Graph the equation of the line.

a $5x - 3y = -15$

b $4x - 3y = -12$

c $2x + y = 10$

d $2x - 3y = 6$

e $5x + y = 1$

4 For each of the following equations of lines:

i Express the equation in slope-intercept form.

ii Determine the slope of the line.

iii State the y -value of the y -intercept of the line.

a $4x - 12y = 4$

b $6x - y = -8$

c $2x + 5y = 35$

5 Consider the slope of a line that passes through the given point:



- i By substituting into the equation $y = mx + b$, find the value of b for this line.
- ii Write the equation of the line in standard form.

a slope = $\frac{7}{3}$, point $(7, -9)$

b slope = 2, point $\left(5, \frac{-9}{5}\right)$.

6 A straight line passes through the point $\left(0, \frac{1}{5}\right)$ with slope 4.

- a Find the equation of the line in the form $y = mx + b$.
- b Express this equation in standard form.
- c Solve for the x -value of the x -intercept.

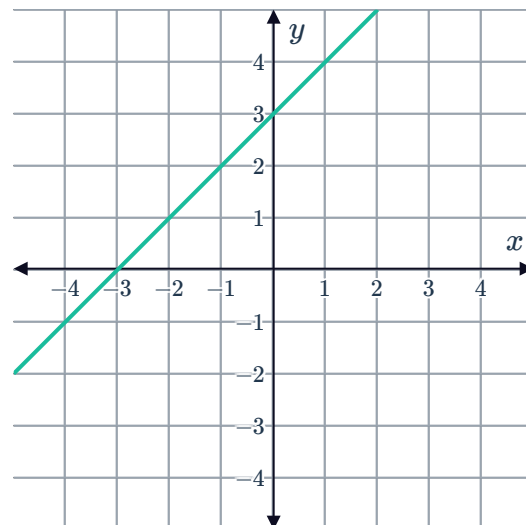
7 Consider the lines with the following equations:

- Line A: $4x + 3y = -5$ and
- Line B: $7x + 9y = 1$

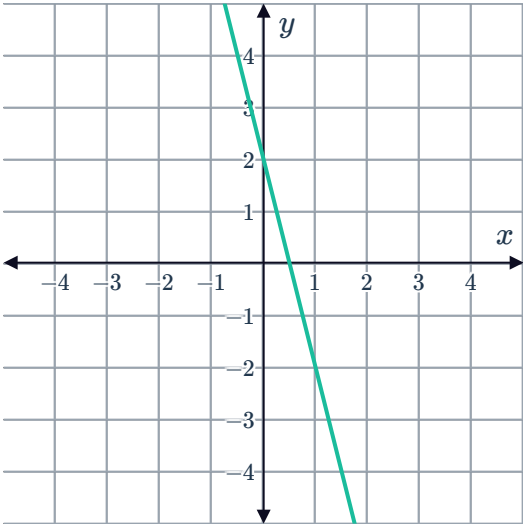
- a Express Line A in slope intercept form:
- b Express Line B in slope intercept form.
- c Determine the slopes of the lines A and B.
- d Identify the steeper line.

8 Which line is steeper, $5x + 5y = -1$ or $5x + 2y = 5$?

9 Write the equation of the graphed line.





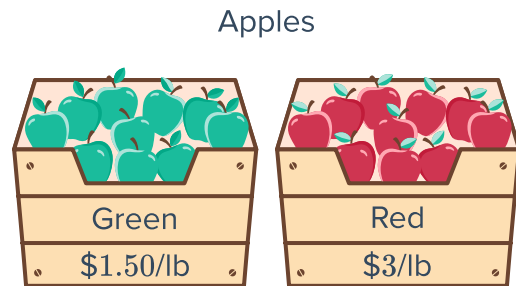
- 10 Determine whether the y -intercept of the straight line is represented by the constant in the following equations.
- a $-5y = 5x - 1$ b $y = -5 + x$ c $-5x - 5y = -5$ d $y = 2x - 2$
- 11 The equation of a straight line is given by $x - 5y = -2$. Solve for the y -value of the y -intercept.
- 12 The point $(-2, -3)$ lies on the line $ax + 3y = 1$.
- a Find the value of a . b Determine the slope of the line.
- 13 Consider the following statement:
Five times the x -value, added to three times the y -value is -4 .
- a Write the statement as an equation in two variables.
b Graph the equation.
- 14 Consider the graph of the line:
- a Determine the y -intercept
b Determine the slope of the line.
c Find the equation of the line in the form $y = mx + b$.
d Express the equation of the line in standard form.
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- 15 The line $kx + 3y = 15$ passes through the point $(13, -8)$.
- a Find the value of k .
b Solve for the x -value of the x -intercept of the line.
c Solve for the y -value of the y -intercept of the line.



Additional questions

- 16 Salvina needs to buy a mix of red and green apples. She is going to spend \$15 in total. Green apples cost \$1.50/lb and red apples cost \$3/lb.

- a Write a linear equation in standard form to represent the scenario. Use the variables x for green and y for red.
- b Calculate and interpret the x -intercept of the linear equation from part (a).
- c Determine the number of green apples she could buy if she bought 2 red apples.
- d State the domain of the linear function.



- 17 Write an expression for the slope, m , of the line represented by the equation $Ax + By = C$. State any restrictions. Explain how this expression might be helpful.
- 18 Describe a scenario where using the standard form of a linear equation is more useful than the slope-intercept form.
- 19 Compare and contrast the standard form and slope-intercept form of a linear equation. Include how they are similar, different and the benefits of each form.
- 20 Effie is an entomologist and is currently studying mosquitos and spiders. She knows that mosquitos have six legs and spiders have eight legs. In her lab, she has a mix of mosquitos and spiders. Between all the bugs, there is a total of 240 legs.
- a Write an equation to represent this scenario.
 - b Draw a graph of the linear equation.
 - c State and describe the domain of the equation.
 - d Explain whether or not every point on the line represents a possible solution.

- 21** Nakisha is selling the baby outfits her daughter has outgrown and is buying second-hand toddler outfits. She is able to sell each baby outfit for \$3 and she buys each toddler outfit for \$5. In the end, she wants to make a profit of exactly \$30.

Profit = revenue – cost

Revenue is the money earned

Cost is the money spent

- a** Write an equation to represent the scenario, using x for the number of baby outfits sold, y for the number of toddler outfits bought, and the profit of exactly \$30.
- b** Calculate and interpret the x -intercept.
- c** Nakisha has 25 baby outfits available to sell and needs 15 toddler outfits. Determine if she is able to make her desired profit of \$30.
- d** Using the additional information from part (c), state and interpret the domain restriction.